

Quiz 1 (Version 3)

CS 320: Concepts of Programming Languages

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- You will have approximately 30 minutes to complete this quiz. Make sure to read every question; some are easier than others.
- Make very clear what your final solution for each problem is (e.g., by surrounding it in a box). We reserve the right to mark off points if we cannot tell what your final solution is.
- You must show your work on all problems unless otherwise specified. A solution without work will be considered incorrect (and will be investigated for potential academic misconduct).
- We will not look at any work on the pages marked *"This page is intentionally left blank."*. You should use these pages for scratch work.

1 OCaml Programming

(10 points) Implement the function

```
val rev_range : int -> int -> int list
```

so that `range i j` is the list of integers from `i` to `j` inclusive in decreasing order. If `i < j` then the range is empty.

```
let _ = assert (range 6 3 = [6;5;4;3])
let _ = assert (range 3 6 = [])
```

You may not use any functions from the standard library except for arithmetic operators and list constructors (i.e., `[]` and `(::)`).

let rec rev_range i j = if i < j then [] else i :: rev_range (i-1) j

2 Types and Values

(10 points) Determine the type of the function `foo` and the value of `bar` below.

```
let rec foo l =  
  match l with  
  | [] -> []  
  | [] :: xs -> foo xs  
  | [x] :: xs -> x :: foo xs  
  | (y :: ys) :: xs -> foo (ys :: xs)  
let bar = foo [[1;2;3];[4;5];[6]]
```

Justify your answers.

`foo` : `int list list` $\rightarrow$ `int list`

technically the type is
`int list list` $\rightarrow$ `int list`

`foo` `[[1;2;3];[4;5];[6]]` =
`foo` `[[2;3];[4;5];[6]]` =
`foo` `[[2];[4;5];[6]]` =
`2 :: foo` `[[4;5];[6]]` =
`2 :: foo` `[[5];[6]]` =
`2 :: 5 :: f` `[[6]]` =
`2 :: 5 :: 6 :: f` `[]` =
`[2; 5; 6]`

Note. This function finds all the last elements
of the inner lists

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